Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Python / LangGraph / GCP & AWS · Healthcare AI & System Architecture · AI Safety & HIPAA Compliance

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SUMMARY

AI Engineer with 9 years of production software ownership. Built AgentProbe -- an open-source AI safety and evaluation platform: guardrails, hallucination detection, output validation, CI-integrated regression gate, and LLM-as-judge scoring with full agent tracing. Built LeaseI -- an 8-agent LangGraph system with RAG, prompt engineering, and HITL on regulated decisions, evaluated across 1,240 test cases at 93.6% precision. Healthcare AI in production: AI on FHIR (HIPAA-compliant, FHIR APIs, PHI, RBAC) and Entrata PHI compliance (zero security incidents across full AWS migration). Led cross-functional engineering delivery across 7 years at enterprise scale. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

AI & LLM Engineering: Python, LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, Large Language Models (LLMs), Generative AI, RAG Pipelines, Vector Databases, ChromaDB, Embeddings, Prompt Engineering, Fine-tuning (HuggingFace Transformers, PyTorch), Tool Calling, Structured Outputs, BERT, spaCy, NLP

AI Safety & Governance: AI Safety Frameworks, Guardrails, Hallucination Detection, Output Validation, HITL (Human-in-the-Loop), LLM Evaluation, LLM-as-Judge, CI/CD Eval Gate, Regression Suites, Agent Tracing, Responsible AI, HIPAA Compliance, PHI Data Handling, RBAC, Data Privacy

Cloud & Infrastructure: Google Cloud Platform (GCP), AWS (S3, EC2, RDS, Glue), Azure, Docker, Kubernetes (K8s), CI/CD (GitHub Actions), FastAPI, REST APIs, RabbitMQ, Redis, Microservices, PostgreSQL, MySQL, Oracle, Git, Linux, Agile/Scrum

Languages & Frameworks: Python, JavaScript, TypeScript, Java, SQL, PL/SQL, React.js, Angular, Flask, Bash, HTML/CSS

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Designed and delivered a production LangGraph agentic system with RAG, prompt-engineered guardrails, and AI safety validation for an external client -- drove the technical roadmap from architecture through delivery, including LLM selection and output validation strategy.

Entrata (PropTech SaaS · 100K+ MAU · PHI compliance · regulated production systems)
Sep 2018 - Aug 2024

Senior Software Engineer

- **Led cross-functional engineering delivery in a regulated domain** -- mentored 5+ engineers, drove technical direction across 120+ Agile sprints, and owned compliance-critical systems (180,000 HUD records, housing eligibility for 100K+ residents); zero compliance violations across 7 years.
- Implemented and owned AI safety and data governance practices: AWS S3 migration with 100% PHI compliance (Libsodium-encrypted pipelines, zero security incidents); designed HIPAA-adjacent data workflows; enforced data privacy and security across all resident data systems.
- Shipped 20+ full-stack production modules (Python, PostgreSQL, JavaScript, RabbitMQ) across Agile sprints; built Python/Bash automation pipelines eliminating 80% of manual workload; optimized 200+ SQL queries (30% performance gain).
- Built internal debugging and tooling infrastructure that reduced investigation time by 40%; conducted 300+ peer code reviews.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Fine-tuned a custom classification model architecture using HuggingFace Transformers and PyTorch; deployed inference API on Quart; presented model design and evaluation results to faculty.
- Built multi-source ML prediction models (Python, R, $R^2 = 0.977$); applied Isolation Forest anomaly detection across 3 research pipelines; delivered Python + database workshops to 600+ students.

PROJECTS

AgentProbe: AI Safety Framework & Evaluation Platform
2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Multi-provider LLM routing

- Open-source AI safety and evaluation platform: guardrails, hallucination detection, output validation, CI-integrated regression gate, LLM-as-judge scoring with full agent tracing, and pass@k harness. Defines and enforces AI safety standards for production agents before and after every deployment.
- 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency); multi-provider routing (Anthropic, OpenAI, Groq, Ollama); 15-endpoint FastAPI backend; evaluated LeaseI across 1,240 test cases.

LeaseI: 8-Agent LangGraph System with RAG & HITL Safety
2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent) · FastAPI · RAG · OCR · ChromaDB · SQLite · Streamlit

- Production agentic AI system: 8-agent LangGraph with RAG, prompt engineering, structured data extraction, and HITL guardrails on regulated decisions. 93.6% precision, 87.9% F1 across 1,240 test cases -- operated safely via AgentProbe AI safety framework.

AI on FHIR: HIPAA-Compliant Healthcare AI System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR APIs · spaCy · BERT · Docker · HIPAA · RBAC

- HIPAA-compliant AI system over structured PHI data -- RBAC access controls, NLP-to-FHIR API translation, LLM + BERT + embeddings integration, data privacy enforced at architecture level. Reduced manual clinical retrieval time ~60%.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017